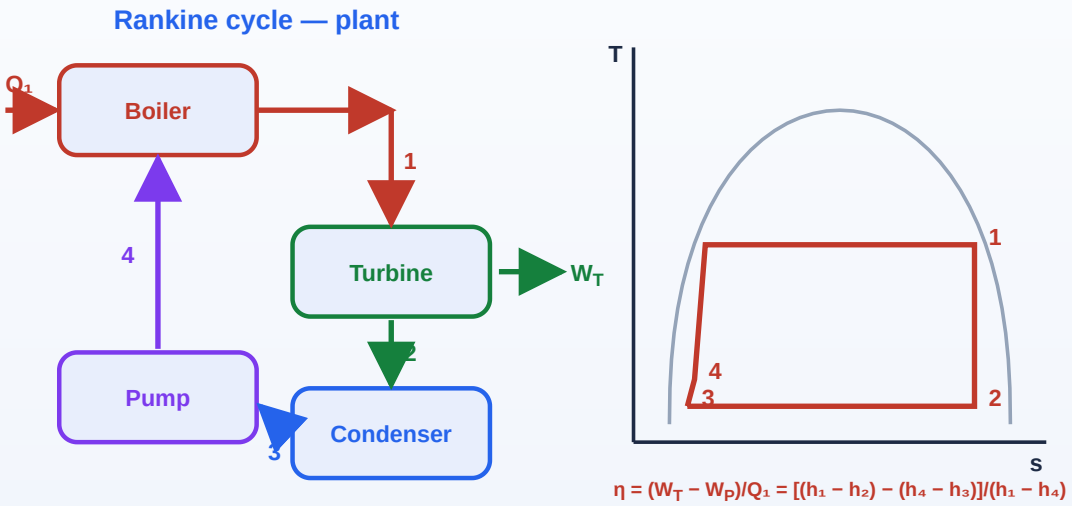
UPSSSC ALERT

UPSSSC JE

MECHANICAL
STEAM & POWER PLANT

Complete English Edition

Unit 6 · 5 Marks · 2026



UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Steam Engineering and Power Plant Engineering**. Syllabus: Rankine cycle efficiency (with and without pump work), boilers, classification, specifications, mountings, accessories, nozzles and steam turbines.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (R. K. Rajput, P. K. Nag, Domkundwar).

For a better score, study this PDF and also attempt the QUIZ.

INDEX

S. No.	Topic	Page No.
1	**Rankine Cycle**	02-09
2	**Boilers — Classification, Specifications, Mountings & Accessories**	10-17
3	**Steam Nozzles**	18-24
4	**Steam Turbines**	25-32

Chapter 1: Rankine Cycle

Rankine Cycle – Basic Concept

The ideal cycle of a steam power plant is the **Rankine cycle**. Water is pumped into the boiler, converted to steam, expanded in the turbine to produce work, then condensed in the condenser back to water, which returns to the pump. Most of India's electricity comes from thermal plants based on this cycle.

1. Key Points

- **Four processes:** 1–2 isentropic expansion in turbine; 2–3 constant-pressure heat rejection in condenser; 3–4 isentropic compression in pump; 4–1 constant-pressure heat addition in boiler.
- **Turbine work:** $W_T = h_1 - h_2$; **pump work:** $W_P = h_4 - h_3 = v_f(p_1 - p_2)$.
- **Efficiency (with pump work):** $\eta = [(h_1 - h_2) - (h_4 - h_3)]/(h_1 - h_4)$.
- **Efficiency (neglecting pump work):** $\eta = (h_1 - h_2)/(h_1 - h_{f2})$ — pump work is very small, so it is often neglected.
- **Work ratio:** net work ÷ turbine work; close to 1 in Rankine (low pump work) — better than Carnot.
- **Specific steam consumption (SSC):** $3600/W_{net}$ kg/kWh.
- **Ways to improve efficiency:** raise boiler pressure, superheat, lower condenser pressure, reheat and regeneration (feed water heaters).
- **Rankine vs Carnot:** compressing wet steam in a Carnot cycle is impractical; Rankine pumps liquid after complete condensation, so it is practical, though less efficient than Carnot.
- **Modified Rankine cycle:** in steam engines expansion is left incomplete (at release), avoiding an excessively long stroke.

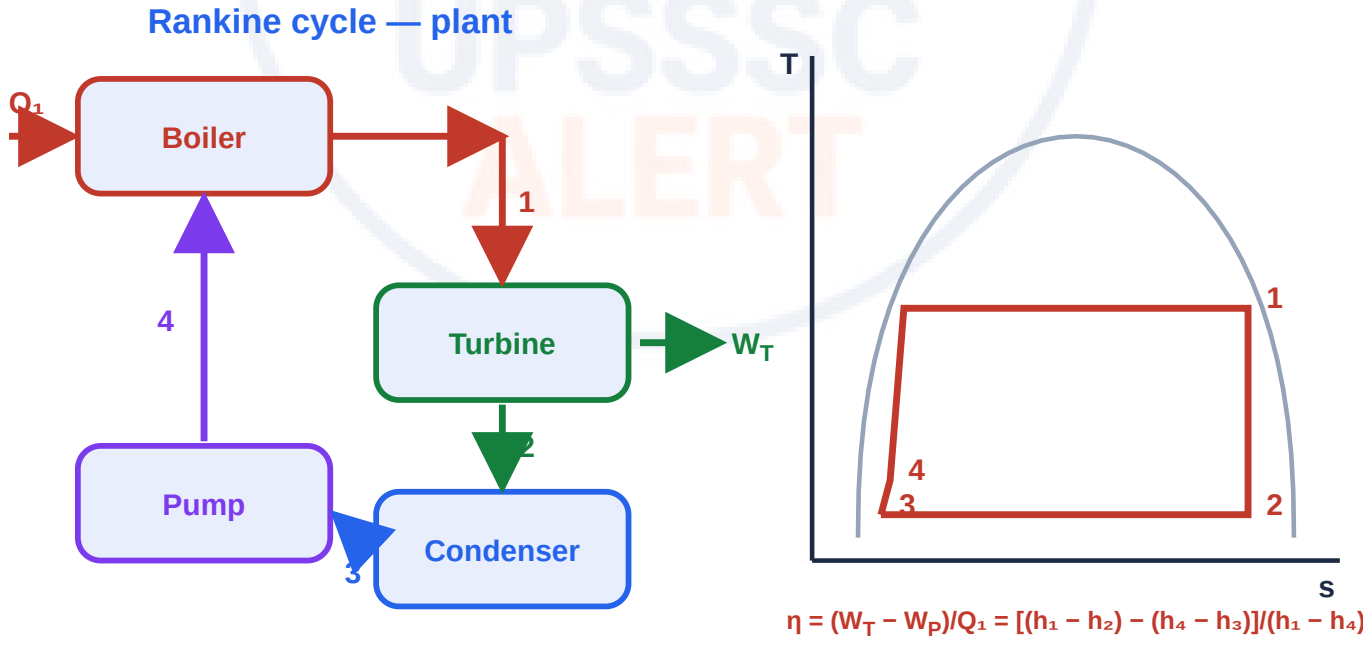


Fig 6.1: Rankine cycle — plant layout and T–s diagram

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Turbine work	$W_T = h_1 - h_2$
Pump work	$W_P = v_f(p_1 - p_2) = h_4 - h_3$
Net work	$W_{net} = W_T - W_P$
Heat supplied	$Q_1 = h_1 - h_4$
Efficiency (with pump)	$\eta = [(h_1 - h_2) - (h_4 - h_3)]/(h_1 - h_4)$
Efficiency (pump neglected)	$\eta = (h_1 - h_2)/(h_1 - h_{f2})$
Work ratio	W_{net}/W_T
Specific steam consumption	$SSC = 3600/W_{net}$ (kg/kWh)
Turbine exit dryness	$s_1 = s_{f2} + x_2\, s_{fg2}$

3. Detailed Explanation

- **Effect of lowering condenser pressure:** mean temperature of heat rejection falls and efficiency rises; but exit steam becomes wetter, causing blade erosion.
- **Superheat and reheat:** superheating raises the mean temperature of heat addition and improves exit dryness; in reheating, partly expanded steam is heated again — mainly to improve dryness.
- **Regenerative cycle:** some steam is bled from the turbine to heat feed water; this raises the mean temperature of heat addition and the efficiency.
- **Turbine exit dryness:** for isentropic expansion $s_1 = s_2$, so $x_2 = (s_1 - s_{f2})/s_{fg2}$; in practice $x_2 \geq 0.88$ is kept.
- **Why pump work is small:** the specific volume of liquid is very small (≈ 0.001 m^3/kg), so $v \cdot \Delta p$ is small.

4. Comparison Table — Rankine vs Carnot Cycle

Basis	Rankine	Carnot
Compression	Pumping of liquid	Compressing wet steam
Practicality	Practical	Impractical
Work ratio	High (≈ 1)	Low
Efficiency (same limits)	Lower	Higher

5. Solved Numerical Problems

Example 1. Rankine cycle: $h_1 = 3200$, $h_2 = 2200$, $h_{f2} = h_3 = 200$ kJ/kg, pump work neglected. Find the efficiency.

Solution:

1. $W_T = 3200 - 2200 = 1000$ kJ/kg; $Q_1 = 3200 - 200 = 3000$ kJ/kg
2. $\eta = 1000/3000 =$ **33.3%**

Example 2. In the above, boiler pressure is 100 bar and condenser pressure 0.1 bar ($v_f = 0.001$ m³/kg). Find efficiency with pump work.

Solution:

1. $W_P = 0.001 \times (10000 - 10) = 9.99 \approx 10$ kJ/kg
2. $h_4 = 210$ kJ/kg; $Q_1 = 3200 - 210 = 2990$ kJ/kg
3. $\eta = (1000 - 10)/2990 =$ **33.1%**

Example 3. Find the specific steam consumption for the above.

Solution:

1. SSC = 3600/990 = **3.64 kg/kWh**

Example 4. Find the work ratio for the above.

Solution:

1. Work ratio = 990/1000 = **0.99**

Example 5. At turbine exit $s_1 = 6.5$, $s_{f2} = 0.65$, $s_{fg2} = 7.5$ kJ/kg·K. Find the dryness.

Solution:

1. $x_2 = (6.5 - 0.65)/7.5 =$ **0.78**

6. Frequently Asked Facts

- Rankine – the ideal cycle of steam power plants.
- Pump work = $v_f \Delta p$, very small; hence work ratio ≈ 1.
- To improve efficiency: high boiler pressure, superheat, low condenser pressure, reheat, regeneration.
- Main purpose of reheating — increase dryness at turbine exit.

7. Note (Confusing Facts)

- Rankine efficiency is lower than Carnot's for the same limits, yet Rankine is practical.
- Lowering condenser pressure raises efficiency but reduces dryness.
- Pump work is $h_4 - h_3$, not $h_3 - h_4$.

Remember: One line: $\eta = W_{net}/Q_1$; SSC = $3600/W_{net}$. Assertion–Reason: (A) The Rankine cycle has a high work ratio. (R) Pumping a liquid needs very little work. — Both true; R explains A.

30 Most Important One-Liner Facts — Rankine Cycle

1. Ideal cycle of steam power plant?

— Rankine.

2. Process in turbine?

— Isentropic expansion.

3. Process in condenser?

— Constant-pressure heat rejection.

4. Process in pump?

— Isentropic compression.

5. Process in boiler?

— Constant-pressure heat addition.

6. Turbine work?

— $h_1 - h_2$.

7. Pump work?

— $v_f(p_1 - p_2)$.

8. Why pump work small?

— Low specific volume of liquid.

9. Rankine efficiency (pump neglected)?

— $(h_1 - h_2)/(h_1 - h_{f2})$.

10. Work ratio?

— Net work ÷ turbine work.

11. Work ratio of Rankine?

— ≈ 1.

12. SSC formula?

— $3600/W_{net}$.

13. Unit of SSC?

— kg/kWh.

14. Efficiency with higher boiler pressure?

— Increases.

15. Efficiency with lower condenser pressure?

— Increases.

16. Drawback of lower condenser pressure?

— Wetter exit steam.

17. Main purpose of reheat?

— Improve exit dryness.

18. In regenerative heating?

— Feed water heated by bled steam.

19. Permissible exit dryness?

— ≥ 0.88.

20. Why Carnot vapour cycle impractical?

— Compression of wet steam.

21. Rankine vs Carnot efficiency?

— Rankine lower.

22. Cycle of steam engine?

— Modified Rankine.

23. Heat supplied (with pump)?

— $h_1 - h_4$.

24. Exit dryness formula?

— $(s_1 - s_{f2})/s_{fg2}$.

25. v_f of water approx.?

— 0.001 m^3/kg.

26. Harm of wet exit steam?

— Blade erosion.

27. Effect of superheating?

— Both efficiency and dryness rise.

28. Components of Rankine plant?

— Boiler, turbine, condenser, pump.

29. Bleeding in?

— Regenerative cycle.

30. When is pump work neglected?

— At low pressures.

Practice MCQs — Rankine Cycle

Q1. Ideal cycle for a steam power plant:

(a) Otto
(b) Diesel
(c) Rankine
(d) Brayton

✔ **Answer: (c) Rankine**

Explanation: Rankine.

Q2. Pump work in the Rankine cycle:

(a) $v_f(p_1 - p_2)$
(b) $h_1 - h_2$
(c) $c_p\Delta T$
(d) Zero

✔ **Answer: (a) $v_f(p_1 - p_2)$**

Explanation: Liquid pumping.

Q3. Rankine efficiency increases with:

(a) Raising condenser pressure
(b) Raising boiler pressure
(c) Reducing superheat
(d) Removing feed heating

✔ **Answer: (b) Raising boiler pressure**

Explanation: Mean temperature of heat addition rises.

Q4. Work ratio of the Rankine cycle is:

(a) Very low
(b) Close to 1
(c) Zero
(d) 2

✔ **Answer: (b) Close to 1**

Explanation: Low pump work.

Q5. $h_1 = 3000$, $h_2 = 2100$, $h_{f2} = 300$ kJ/kg (pump neglected) — efficiency:

(a) 30%
(b) 33.3%
(c) 40%
(d) 70%

✔ **Answer: (b) 33.3%**

Explanation: 900/2700.

Q6. Specific steam consumption =

(a) $W_{net}/3600$
(b) $3600/W_{net}$
(c) $3600 \times W_{net}$
(d) Q_1/W_{net}

✔ **Answer: (b) $3600/W_{net}$**

Explanation: kg/kWh.

Q7. Main purpose of reheating:

(a) Greatly raise efficiency
(b) Increase turbine exit dryness
(c) Reduce pump work
(d) Lower boiler pressure

✔ **Answer: (b) Increase turbine exit dryness**

Explanation: Prevent erosion.

Q8. Expansion in the turbine of a Rankine cycle is assumed:

(a) Isothermal
(b) Isentropic
(c) Isobaric
(d) Throttling

✔ **Answer: (b) Isentropic**

Explanation: Reversible adiabatic.

Q9. In a regenerative cycle feed water is heated by:

(a) Exhaust gas
(b) Steam bled from turbine
(c) Electric heater
(d) Solar energy

✔ **Answer: (b) Steam bled from turbine**

Explanation: Bleeding.

Q10. Lowering condenser pressure makes exit steam:

(a) Drier
(b) Wetter
(c) Superheated
(d) Unchanged

✔ **Answer: (b) Wetter**

Explanation: More expansion.

Q11. Rankine compared to Carnot (same limits):

(a) More efficient
(b) Less efficient, more practical
(c) Equal
(d) Impractical

✔ **Answer: (b) Less efficient, more practical**

Explanation: No wet-steam compression.

Q12. W_{net} = 1000 kJ/kg — SSC:

(a) 1 kg/kWh
(b) 3.6 kg/kWh
(c) 0.36 kg/kWh
(d) 36 kg/kWh

✔ **Answer: (b) 3.6 kg/kWh**

Explanation: 3600/1000.

Q13. Heat addition in the boiler is at constant:

(a) Volume
(b) Pressure
(c) Temperature
(d) Entropy

✔ **Answer: (b) Pressure**

Explanation: Constant pressure.

Q14. Pump work is usually ____ of turbine work:

(a) 1–2%
(b) 20%
(c) 50%
(d) 90%

✔ **Answer: (a) 1–2%**

Explanation: Very small.

Q15. $s_1 = 6.6$, $s_{f2} = 0.6$, $s_{fg2} = 7.5$ — x_2:

(a) 0.70
(b) 0.80
(c) 0.88
(d) 0.95

✔ **Answer: (b) 0.80**

Explanation: 6/7.5.

SSC JE / UPSSSC JE Previous Year Questions — Rankine Cycle

Q1. The Rankine cycle consists of:

SSC JE (Previous Year)

(a) Two isentropic and two constant pressure processes
(b) Two isothermal and two isentropic
(c) Two constant volume and two isentropic
(d) Four isothermal

✔ **Answer: (a) Two isentropic and two constant pressure processes**

Explanation: Rankine.

Q2. Rankine efficiency can be increased by:

SSC JE (Previous Year)

(a) Decreasing condenser pressure
(b) Decreasing boiler pressure
(c) Decreasing superheat
(d) All of these

✔ **Answer: (a) Decreasing condenser pressure**

Explanation: Lower condenser pressure.

Q3. Regenerative heating:

SSC JE (Previous Year)

(a) Increases efficiency
(b) Decreases efficiency
(c) Increases work
(d) Has no effect

✔ **Answer: (a) Increases efficiency**

Explanation: Mean heat-addition temperature rises.

Q4. Pump work in Rankine is negligible because:

SSC JE (Previous Year)

(a) Low specific volume of liquid
(b) Small pressure difference
(c) Efficient pump
(d) High temperature

✔ **Answer: (a) Low specific volume of liquid**

Explanation: v_f is very small.

Q5. In a reheat cycle steam is reheated:

SSC JE (Previous Year)

(a) Before boiler

(b) After partial expansion in turbine

(c) After condenser

(d) After pump

✔ **Answer: (b) After partial expansion in turbine**

Explanation: After the HP turbine.

UPSSSC ALERT

Chapter 2: Boilers — Classification, Specifications, Mountings & Accessories

Boilers — Classification, Specifications, Mountings & Accessories – Basic Concept

A **boiler (steam generator)** is a closed vessel in which water is converted into steam using the heat of combustion of fuel. Under the Indian Boilers Act 1923, a closed vessel of capacity over 22.75 litres generating steam above 1 kg/cm² (gauge) is a boiler. Boilers are used for power generation, industry (textile, sugar, paper) and heating.

1. Key Points

- **What flows in tubes:** fire-tube — hot gases inside tubes, water outside (Cornish, Lancashire, Cochran, Locomotive); water-tube — water inside tubes, gases outside (Babcock & Wilcox, Stirling, La Mont, Benson).
- **Axis of shell:** horizontal (Lancashire, Locomotive), vertical (Cochran, simple vertical) and inclined.
- **Furnace position:** internally fired (Lancashire, Cochran) and externally fired (Babcock & Wilcox).
- **Water circulation:** natural (density difference — B&W) and forced (pump — La Mont, Benson, Velox).
- **Pressure:** low pressure (< 10 bar – Cochran, Lancashire), high pressure (B&W, La Mont), supercritical (> 221 bar — Benson; drumless).
- **Boiler mountings (for safety, compulsory):** safety valves (2), water level indicators (2), pressure gauge, fusible plug, steam stop valve, feed check valve, blow-off cock, manhole.
- **Boiler accessories (to raise efficiency):** economiser (heats feed water by flue gas), air preheater (heats combustion air), superheater (superheats steam), feed pump, injector, steam separator, steam trap.
- **Types of safety valve:** dead weight, lever, spring loaded (locomotive/marine), and high steam–low water safety valve.
- **Fusible plug:** on the furnace crown; when water level falls, the low-melting metal melts, letting steam and water into the furnace to put out the fire.
- **Boiler specifications:** type, working pressure, steam generation capacity (kg/h), equivalent evaporation, heating surface area, efficiency, type of fuel.

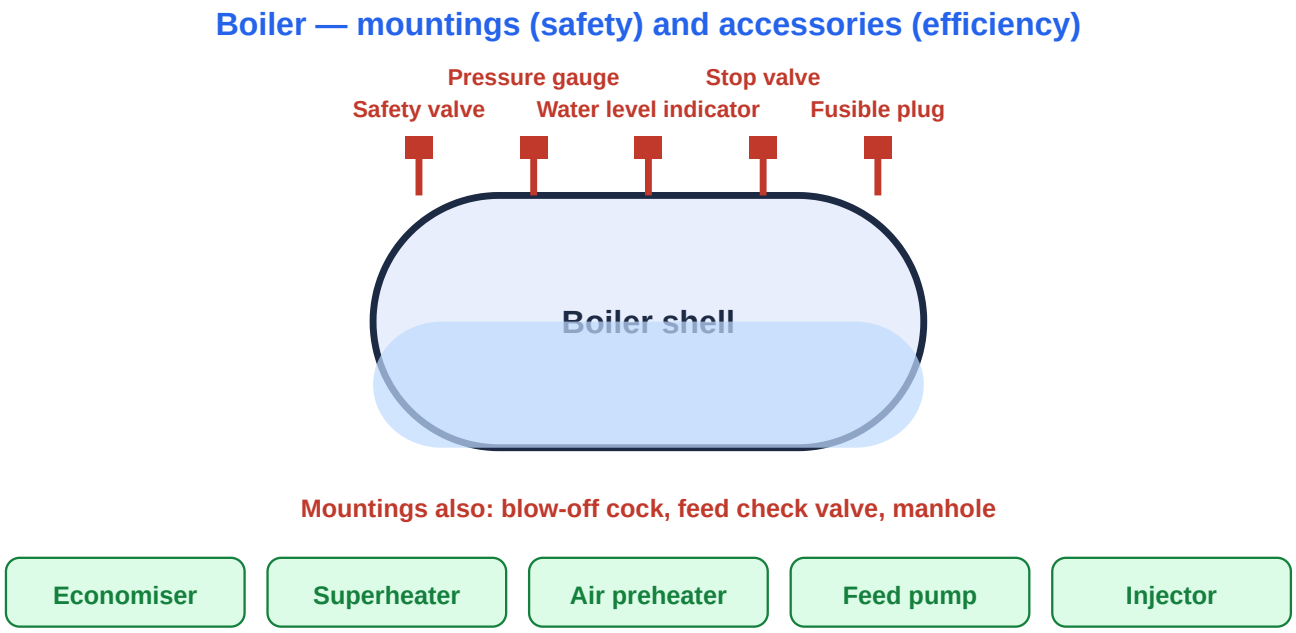


Fig 6.2: Boiler mountings and accessories

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Boiler efficiency	$\eta = m_s(h - h_{fw}) / (m_f \times CV)$
Equivalent evaporation	$m_e = m_s(h - h_{fw})/2257$
Factor of evaporation	$F = (h - h_{fw})/2257$
Actual evaporation	$m_a = m_s/m_f$ (kg steam/kg fuel)
Latent heat 'from and at 100°C'	2257 kJ/kg
Natural draught of chimney	$h_w = 353H(1/T_a - (m+1)/(m\ T_g))$ mm

3. Detailed Explanation

- **Cochran boiler:** vertical, multi-fire-tube, internally fired; compact, ~10 bar; for small industries.
- **Lancashire boiler:** horizontal, two large fire tubes, internally fired; large steam storage, handles sudden load changes; used in sugar mills.
- **Babcock & Wilcox boiler:** horizontal inclined water tubes, externally fired, natural circulation; high pressure and capacity — used in power plants.
- **La Mont and Benson:** forced circulation; Benson works drumless at supercritical pressure — water flashes directly into steam.
- **Economiser vs air preheater:** both recover flue-gas heat — the first heats feed water, the second heats combustion air; order: boiler → superheater → economiser → air preheater → chimney.
- **Draught:** the pressure difference that drives air through the furnace; natural (chimney) and artificial — induced (fan at chimney base), forced (fan before furnace), balanced.

4. Comparison Table — Fire-tube vs Water-tube Boiler

Basis	Fire-tube	Water-tube
Inside tubes	Hot gases	Water
Pressure limit	Low (up to ≈ 20 bar)	High (> 100 bar)
Steam rate	Low	High
Explosion risk	More (large water volume)	Less
Examples	Cochran, Lancashire, Locomotive	B&W, La Mont, Benson

4. Comparison Table — Mountings vs Accessories

Basis	Mountings	Accessories
Purpose	Safety and control	Increase efficiency
Compulsory	Yes (IBR)	No
Location	On the boiler	Separate units connected
Examples	Safety valve, fusible plug	Economiser, superheater

5. Solved Numerical Problems

Example 1. A boiler produces 80 kg of steam from 10 kg of fuel (CV = 30,000 kJ/kg); h = 2800, h_{fw} = 200 kJ/kg. Find the efficiency.

Solution:

1. $\eta = 80 \times (2800 - 200)/(10 \times 30000) = 208000/300000 =$ **69.3%**

Example 2. For the above, find the equivalent evaporation per kg of fuel.

Solution:

1. $m_a = 80/10 = 8$ kg/kg
2. $m_e = 8 \times 2600/2257 =$ **9.22 kg/kg of fuel**

Example 3. Find the factor of evaporation if h = 2900 and h_{fw} = 300 kJ/kg.

Solution:

1. $F = (2900 - 300)/2257 =$ **1.152**

Example 4. A boiler generates 5000 kg/h of steam ($h - h_{fw}$ = 2500 kJ/kg) at 80% efficiency. Find fuel (CV = 40,000) consumption.

Solution:

1. $m_f = 5000 \times 2500/(0.8 \times 40000) =$ **390.6 kg/h**

6. Frequently Asked Facts

- Fire-tube: Cochran, Lancashire, Locomotive, Cornish; water-tube: B&W, Stirling, La Mont, Benson.

- Benson — supercritical, drumless.
- Mountings for safety, accessories for efficiency.
- Equivalent evaporation 'from and at 100°C' — 2257 kJ/kg.

7. Note (Confusing Facts)

- Economiser heats feed water; air preheater heats air — both are accessories.
- Feed check valve is a mounting; feed pump is an accessory.
- A locomotive boiler is fire-tube, though it gives a high steam rate.

Remember: One line: 2 safety valves + 2 water level indicators are compulsory. Assertion–Reason: (A) Water-tube boilers suit high pressure. (R) Small-diameter tubes withstand higher pressure. — Both true; R explains A.

30 Most Important One-Liner Facts — Boilers — Classification, Specifications, Mountings & Accessories

1. Gases inside tubes — boiler?

— Fire-tube.

2. Water inside tubes — boiler?

— Water-tube.

3. Type of Cochran boiler?

— Vertical, multi-fire-tube.

4. Fire tubes in Lancashire?

— Two.

5. Fire tubes in Cornish?

— One.

6. Type of B&W boiler?

— Water-tube, externally fired.

7. Forced circulation boilers?

— La Mont, Benson, Velox.

8. Drumless boiler?

— Benson.

9. Supercritical boiler?

— Benson.

10. Locomotive boiler type?

— Horizontal fire-tube.

11. Purpose of mountings?

— Safety.

12. Purpose of accessories?

— Increase efficiency.

13. Compulsory safety valves?

— 2.

14. Compulsory water gauges?

— 2.

15. Puts off fire on low water?

— Fusible plug.

16. Location of fusible plug?

— Furnace crown.

17. Heats feed water with flue gas?

— Economiser.

18. Heats combustion air?

— Air preheater.

19. Superheats steam?

— Superheater.

20. Removes water and sludge?

— Blow-off cock.

21. Prevents water flowing back?

— Feed check valve.

22. Safety valve on locomotives?

— Spring loaded.

23. Boiler efficiency?

— $m_s(h - h_{fw})/(m_f \text{ CV})$.

24. Latent heat in equivalent evaporation?

— 2257 kJ/kg.

25. Year of Indian Boilers Act?

— 1923.

26. Fan at chimney base — draught?

— Induced.

27. Fan before furnace — draught?

— Forced.

28. Draught by chimney?

— Natural.

29. Removes condensate?

— Steam trap.

30. Feeds water without pump?

— Injector.

Practice MCQs — Boilers — Classification, Specifications, Mountings & Accessories

Q1. Which is a water-tube boiler?

(a) Cochran

(b) Lancashire

(c) Babcock & Wilcox

(d) Locomotive

✔ **Answer: (c) Babcock & Wilcox**

Explanation: B&W is water-tube.

Q2. Function of the fusible plug:

(a) Measure pressure

(b) Extinguish fire on low water

(c) Superheat steam

(d) Heat feed water

✔ **Answer: (b) Extinguish fire on low water**

Explanation: Safety mounting.

Q3. An economiser heats:

(a) Combustion air

(b) Feed water

(c) Steam

(d) Fuel

✔ **Answer: (b) Feed water**

Explanation: Using flue-gas heat.

Q4. Which is a boiler accessory?

(a) Safety valve

(b) Pressure gauge

(c) Superheater

(d) Fusible plug

✔ **Answer: (c) Superheater**

Explanation: For efficiency.

Q5. Drumless boiler working at supercritical pressure:

(a) La Mont

(b) Benson

(c) Lancashire

(d) Cochran

✔ **Answer: (b) Benson**

Explanation: Benson.

Q6. Number of fire tubes in a Lancashire boiler:

(a) 1
(b) 2
(c) 4
(d) Many

✔ **Answer: (b) 2**

Explanation: Two.

Q7. A locomotive boiler is:

(a) Water-tube
(b) Horizontal multi-fire-tube
(c) Vertical water-tube
(d) Forced circulation

✔ **Answer: (b) Horizontal multi-fire-tube**

Explanation: Fire-tube.

Q8. Number of compulsory safety valves on a boiler:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

Explanation: IBR.

Q9. Device that heats combustion air:

(a) Economiser
(b) Air preheater
(c) Superheater
(d) Condenser

✔ **Answer: (b) Air preheater**

Explanation: From flue gas.

Q10. To remove sludge and sediment from a boiler:

(a) Blow-off cock
(b) Stop valve
(c) Feed check valve
(d) Safety valve

✔ **Answer: (a) Blow-off cock**

Explanation: Blow-off cock.

Q11. 80 kg steam/10 kg fuel, $h - h_{fw} = 2257$ kJ/kg — equivalent evaporation:

(a) 8 kg/kg
(b) 10 kg/kg
(c) 2257 kg/kg
(d) 0.8 kg/kg

✔ **Answer: (a) 8 kg/kg**

Explanation: F = 1.

Q12. Natural circulation is due to:

(a) Pump
(b) Density difference
(c) Chimney
(d) Fan

✔ **Answer: (b) Density difference**

Explanation: Hot water is lighter.

Q13. Low-pressure vertical boiler for small industries:

(a) Cochran
(b) B&W
(c) Benson
(d) La Mont

✔ **Answer: (a) Cochran**

Explanation: Cochran.

Q14. Fan at the base of the chimney — draught is:

(a) Natural
(b) Forced
(c) Induced
(d) Balanced

✔ **Answer: (c) Induced**

Explanation: It sucks gases.

Q15. The feed check valve is a:

(a) Mounting
(b) Accessory
(c) Fuel valve
(d) Turbine part

✔ **Answer: (a) Mounting**

Explanation: Compulsory fitting on boiler.

SSC JE / UPSSSC JE Previous Year Questions — Boilers — Classification, Specifications, Mountings & Accessories

Q1. Which is not a boiler mounting?

SSC JE (Previous Year)

(a) Safety valve
(b) Water level indicator
(c) Economiser
(d) Fusible plug

✔ **Answer: (c) Economiser**

Explanation: Economiser is an accessory.

Q2. Cochran boiler is a:

SSC JE (Previous Year)

(a) Horizontal water-tube
(b) Vertical fire-tube
(c) Horizontal fire-tube
(d) Vertical water-tube

✔ **Answer: (b) Vertical fire-tube**

Explanation: Vertical multi-fire-tube.

Q3. Special feature of Benson boiler:

SSC JE (Previous Year)

(a) Large drum
(b) No drum
(c) Low pressure
(d) Fire-tube

✔ **Answer: (b) No drum**

Explanation: Supercritical, drumless.

Q4. Function of a superheater:

SSC JE (Previous Year)

(a) Heat feed water
(b) Raise steam temperature above saturation
(c) Heat air
(d) Increase pressure

✔ **Answer: (b) Raise steam temperature above saturation**

Explanation: Superheating.

Q5. Lancashire boiler is:

SSC JE (Previous Year)

(a) Internally fired horizontal fire-tube
(b) Externally fired water-tube
(c) Vertical
(d) Forced circulation

✔ **Answer: (a) Internally fired horizontal fire-tube**

Explanation: Two fire tubes.

Q6. Protects the boiler when water level falls:

SSC JE (Previous Year)

(a) Fusible plug
(b) Blow-off cock
(c) Stop valve
(d) Injector

✔ **Answer: (a) Fusible plug**

Explanation: Plug melts and quenches the fire.

UPSSSC ALERT

Chapter 3: Steam Nozzles

Steam Nozzles – Basic Concept

A **nozzle** is a passage of varying cross-section in which steam pressure drops and its enthalpy is converted into kinetic energy (high velocity). In steam turbines the high-velocity jet from nozzles strikes the blades and rotates them. Nozzles are also used in injectors and ejectors.

1. Key Points

- **Types of nozzles:** convergent (up to sonic velocity), divergent, and convergent–divergent (C-D / de Laval – for supersonic velocity).
- **Exit velocity (from SFEE):** $C_2 = \sqrt{[2(h_1 - h_2) + C_1^2]} \approx 44.72\sqrt{(h_1 - h_2)}$ m/s (h in kJ/kg, C_1 negligible).
- **Throat:** the section of minimum area in a C-D nozzle; at maximum flow the velocity there is sonic (M = 1).
- **Critical pressure ratio:** $p_c/p_1 = [2/(n + 1)]^{n/(n-1)}$; superheated steam (n = 1.3) 0.546, saturated steam (n = 1.135) 0.577, air (n = 1.4) 0.528.
- **Choked flow:** reducing the back pressure below critical does not increase mass flow in a convergent nozzle; it stays at its maximum.
- **Nozzle efficiency:** η_n = actual enthalpy drop ÷ isentropic enthalpy drop; friction reduces the actual velocity. Velocity coefficient = $\sqrt{\eta_n}$.
- **Supersaturated (metastable) flow:** in rapid expansion steam crosses the saturation line without condensing for a while, up to the Wilson line. This slightly increases mass flow.
- **Mass flow rate:** m = A·C/v, where v is the specific volume of steam at that section.

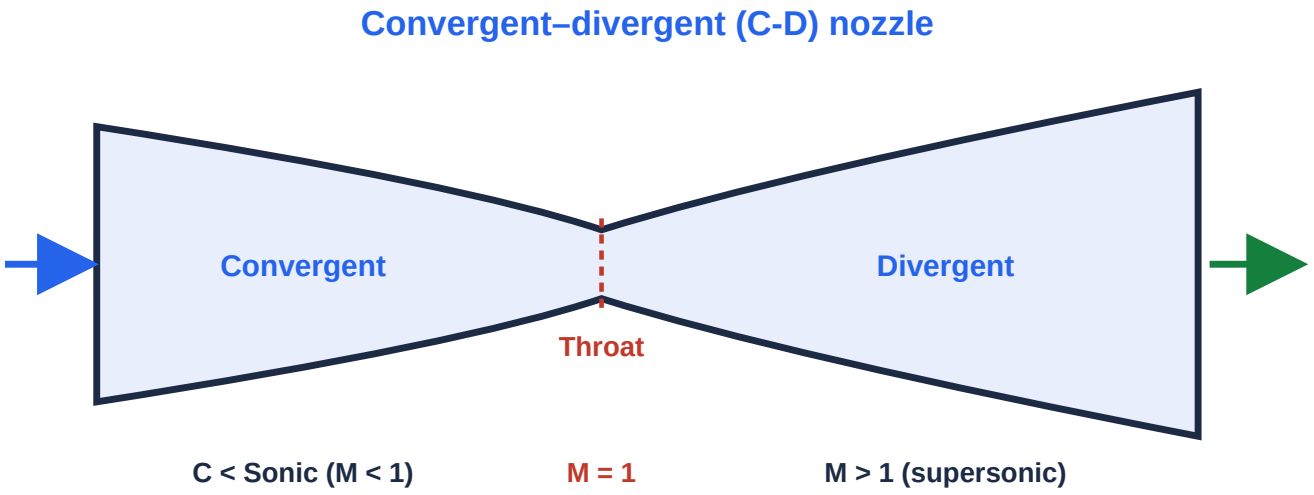


Fig 6.3: Steam nozzle — convergent–divergent

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Exit velocity	$C_2 = 44.72\sqrt{(h_1 - h_2)}$
With inlet velocity	$C_2 = \sqrt{[2000(h_1 - h_2) + C_1^2]}$
Critical pressure ratio	$[2/(n+1)]^{n/(n-1)}$
Superheated steam	$n = 1.3 \rightarrow 0.546$
Saturated steam	$n = 1.135 \rightarrow 0.577$
Air	$n = 1.4 \rightarrow 0.528$
Mass flow	$m = A\,C / v$
Nozzle efficiency	$\eta_n = \Delta h_{actual}/\Delta h_{isentropic}$
Velocity coefficient	$C_{actual}/C_{isentropic} = \sqrt{\eta_n}$

3. Detailed Explanation

- **Area–velocity relation:** $dA/A = (M^2 - 1)\,dC/C$. In subsonic flow (M < 1) area must decrease to accelerate (convergent); in supersonic flow (M > 1) area must increase (divergent).
- **Use of C-D nozzle:** when the exit pressure is below the critical pressure (large pressure ratio), for supersonic velocity; e.g. de Laval turbines and rockets.
- **Effect of friction in nozzle:** actual enthalpy drop decreases, exit steam is drier (reheat effect), entropy increases.
- **Index n of steam expansion:** $pv^n = C$; superheated 1.3, saturated 1.135, wet steam n = 1.035 + 0.1x (Zeuner).

4. Comparison Table — Convergent vs C-D Nozzle

Basis	Convergent	Convergent–divergent
Max exit velocity	Sonic (M = 1)	Supersonic (M > 1)
Use	Small pressure ratio	Large pressure ratio
Throat	Exit is the throat	Throat in the middle
Example	Impulse stages	De Laval turbine, rocket

5. Solved Numerical Problems

Example 1. Isentropic enthalpy drop in a steam nozzle is 200 kJ/kg, inlet velocity negligible. Find the exit velocity.

Solution:

1. $C_2 = 44.72 \sqrt{200} = 44.72 \times 14.14 =$ **632.5 m/s**

Example 2. If the above nozzle efficiency is 90%, find the actual velocity.

Solution:

1. $\Delta h = 0.9 \times 200 = 180$ kJ/kg
2. $C = 44.72 \sqrt{180} =$ **600 m/s**

Example 3. Superheated steam enters a nozzle at 10 bar. Find the throat pressure.

Solution:

1. $p_c = 0.546 \times 10 =$ **5.46 bar**

Example 4. At exit C = 500 m/s, v = 0.5 m³/kg and flow 2 kg/s. Find the exit area.

Solution:

1. $A = m\,v/C = 2 \times 0.5/500 = 0.002$ m² = **2000 mm²**

6. Frequently Asked Facts

- $C = 44.72\sqrt{\Delta h}$ — the most asked formula.
- Critical pressure ratio: superheated 0.546, saturated 0.577, air 0.528.
- M = 1 at the throat; C-D nozzle for supersonic flow.
- Friction in a nozzle makes exit steam drier.

7. Note (Confusing Facts)

- In subsonic flow the convergent part accelerates; in supersonic flow the divergent part does.
- After choking, lowering back pressure does not increase flow.
- The factor 44.72 applies only when Δh is in kJ/kg.

Remember: One line: nozzle – enthalpy → kinetic energy; diffuser – the reverse. Assertion–Reason: (A) A convergent nozzle cannot give supersonic velocity. (R) The maximum velocity at the throat is

sonic. — Both true; R explains A.

30 Most Important One-Liner Facts — Steam Nozzles

1. Function of nozzle?

— Convert enthalpy into KE.

2. Function of diffuser?

— Convert KE into pressure.

3. Exit velocity formula?

— $44.72\sqrt{\Delta h}$.

4. Nozzle for supersonic?

— Convergent–divergent.

5. Other name of C-D nozzle?

— De Laval nozzle.

6. Section of minimum area?

— Throat.

7. Mach number at throat?

— 1.

8. Critical ratio for superheated?

— 0.546.

9. Critical ratio for saturated?

— 0.577.

10. Critical ratio for air?

— 0.528.

11. n for superheated steam?

— 1.3.

12. n for saturated steam?

— 1.135.

13. n for wet steam (Zeuner)?

— 1.035 + 0.1x.

14. Mass flow?

— AC/v.

15. Nozzle efficiency?

— Actual ÷ isentropic enthalpy drop.

16. Velocity coefficient?

— $\sqrt{\eta_n}$.

17. Area to accelerate subsonic flow?

— Decrease.

18. Area to accelerate supersonic flow?

— Increase.

19. Condition of maximum flow?

— Choked.

20. Limit of supersaturated flow?

— Wilson line.

21. Mass flow with supersaturation?

— Increases slightly.

22. Exit steam with friction?

— Drier.

23. Entropy with friction?

— Increases.

24. Area–velocity relation?

— $dA/A = (M^2 - 1)dC/C$.

25. Max velocity in convergent nozzle?

— Sonic velocity.

26. Flow in nozzle assumed?

— Adiabatic.

27. Work in nozzle?

— Zero.

28. Velocity for Δh = 125 kJ/kg?

— 500 m/s.

29. Nozzle in rocket?

— Convergent–divergent.

30. Throat pressure for 10 bar superheated?

— 5.46 bar.

Practice MCQs — Steam Nozzles

Q1. Exit velocity of steam in a nozzle (Δh in kJ/kg):

(a) 4.472√Δh
(b) 44.72√Δh
(c) 447.2√Δh
(d) 91.5√Δh

✔ **Answer: (b) 44.72√Δh**

Explanation: √(2000Δh).

Q2. Critical pressure ratio for superheated steam:

(a) 0.528
(b) 0.546
(c) 0.577
(d) 0.6

✔ **Answer: (b) 0.546**

Explanation: n = 1.3.

Q3. Nozzle used for supersonic velocity:

(a) Convergent
(b) Divergent
(c) Convergent–divergent
(d) Parallel

✔ **Answer: (c) Convergent–divergent**

Explanation: Divergent after throat.

Q4. Mach number at the throat (maximum flow):

(a) 0.5
(b) 1
(c) 2
(d) Zero

✔ **Answer: (b) 1**

Explanation: Sonic.

Q5. Value of n for saturated steam:

(a) 1.3
(b) 1.4
(c) 1.135
(d) 1.035

✔ **Answer: (c) 1.135**

Explanation: 1.135.

Q6. Δh = 180 kJ/kg — exit velocity:

(a) 420 m/s
(b) 600 m/s
(c) 800 m/s
(d) 180 m/s

✔ **Answer: (b) 600 m/s**

Explanation: 44.72 × 13.42.

Q7. Friction in a nozzle makes exit steam:

(a) Wetter
(b) Drier
(c) Unchanged
(d) Subcooled

✔ **Answer: (b) Drier**

Explanation: Reheat effect.

Q8. To accelerate subsonic flow, the nozzle must be:

(a) Convergent
(b) Divergent
(c) Parallel
(d) Any

✔ **Answer: (a) Convergent**

Explanation: $dA < 0$.

Q9. After choking, reducing back pressure makes the flow:

(a) Increase
(b) Decrease
(c) Remain constant
(d) Zero

✔ **Answer: (c) Remain constant**

Explanation: Maximum flow.

Q10. Limit of supersaturated flow:

(a) Wilson line
(b) Saturation line
(c) Critical point
(d) Fanno line

✔ **Answer: (a) Wilson line**

Explanation: Wilson line.

Q11. Nozzle efficiency 81% — velocity coefficient:

(a) 0.81
(b) 0.9
(c) 0.66
(d) 0.95

✔ **Answer: (b) 0.9**

Explanation: $\sqrt{0.81}$.

Q12. A de Laval nozzle is:

(a) Convergent
(b) Convergent–divergent
(c) Divergent
(d) Diffuser

✔ **Answer: (b) Convergent–divergent**

Explanation: C-D.

Q13. Steam flow in a nozzle is assumed:

(a) Isothermal
(b) Adiabatic
(c) Constant pressure
(d) Constant volume

✔ **Answer: (b) Adiabatic**

Explanation: Negligible heat transfer.

Q14. Critical pressure ratio for air:

(a) 0.528
(b) 0.546
(c) 0.577
(d) 0.5

✔ **Answer: (a) 0.528**

Explanation: n = 1.4.

Q15. 2 kg/s, v = 0.5 m³/kg, C = 500 m/s — exit area:

(a) 200 mm²
(b) 2000 mm²
(c) 20000 mm²
(d) 500 mm²

✔ **Answer: (b) 2000 mm²**

Explanation: mv/C = 0.002 m².

SSC JE / UPSSSC JE Previous Year Questions — Steam Nozzles

Q1. Purpose of a steam nozzle:

SSC JE (Previous Year)

(a) Increase pressure
(b) Convert heat energy into kinetic energy
(c) Condense steam
(d) Superheat steam

✔ **Answer: (b) Convert heat energy into kinetic energy**

Explanation: Enthalpy → velocity.

Q2. Critical pressure ratio for dry saturated steam:

SSC JE (Previous Year)

(a) 0.528
(b) 0.546
(c) 0.577
(d) 0.582

✔ **Answer: (c) 0.577**

Explanation: 0.577.

Q3. Velocity in the divergent part of a C-D nozzle:

SSC JE (Previous Year)

(a) Decreases
(b) Increases (supersonic)
(c) Remains constant
(d) Zero

✔ **Answer: (b) Increases (supersonic)**

Explanation: M > 1.

Q4. Exit velocity of steam from a nozzle depends on:

SSC JE (Previous Year)

(a) Enthalpy drop
(b) Nozzle length
(c) Colour of steam
(d) Mass

✔ **Answer: (a) Enthalpy drop**

Explanation: C ∝ √Δh.

Chapter 4: Steam Turbines

Steam Turbines – Basic Concept

A **steam turbine** is a rotary machine in which the heat energy of steam is first converted into kinetic energy in nozzles/blades and then into rotary mechanical work. It is the main prime mover of thermal and nuclear power plants, with large capacity, high speed and balanced motion.

1. Key Points

- **Impulse turbine:** the entire pressure drop occurs in nozzles only; pressure stays constant over moving blades, only velocity falls. Examples: De Laval (simplest, single stage), Curtis, Rateau.
- **Reaction turbine:** pressure drop in both fixed and moving blades; moving blades also act as nozzles. Example: Parsons (50% reaction).
- **Degree of reaction (R):** enthalpy drop in moving blades ÷ enthalpy drop in the whole stage. Impulse R = 0; Parsons R = 0.5.
- **Compounding:** to reduce the excessive speed of a single stage (De Laval ~30,000 rpm) — velocity compounding (Curtis), pressure compounding (Rateau), pressure–velocity compounding.
- **Blade efficiency (impulse):** maximum when blade speed ratio $\rho = u/C_1 = \cos \alpha/2$; then $\eta_{max} = \cos^2\alpha$ (symmetric blades, frictionless).
- **Parsons turbine:** maximum efficiency at $u/C_1 = \cos \alpha$; $\eta_{max} = 2\cos^2\alpha/(1 + \cos^2\alpha)$.
- **Velocity diagram:** absolute velocity C, blade velocity u, relative velocity V; work from whirl components: $W = u(C_{w1} \pm C_{w2})$ per kg.
- **Turbine governing:** throttle governing, nozzle control governing, bypass governing.
- **By flow:** axial (common) and radial (Ljungström). By exhaust: condensing and back-pressure turbines.

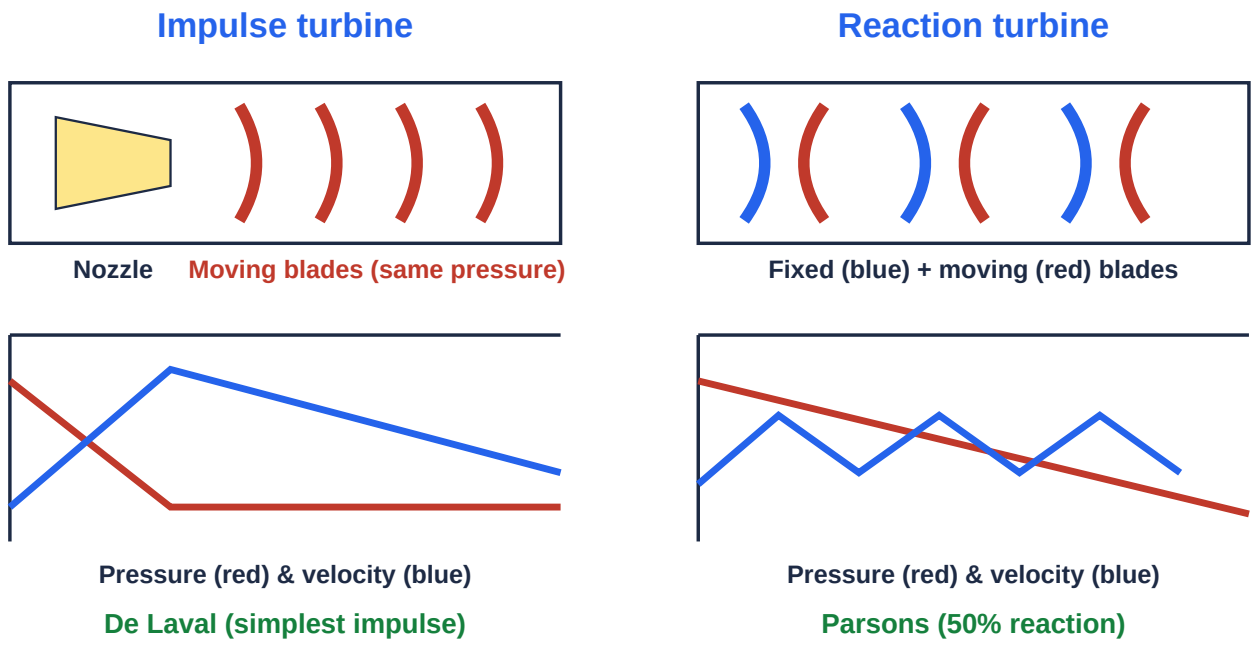


Fig 6.4: Pressure and velocity variation in impulse and reaction turbines

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Work per kg	$W = u\,(C_{w1} \pm C_{w2})$
Blade efficiency	$\eta_b = 2u(C_{w1} \pm C_{w2})/C_1^2$
Impulse: optimum speed ratio	$u/C_1 = \cos\alpha / 2$
Impulse: max efficiency	$\cos^2\alpha$
Parsons: optimum ratio	$u/C_1 = \cos\alpha$
Parsons: max efficiency	$2\cos^2\alpha/(1 + \cos^2\alpha)$
Degree of reaction	$R = \Delta h_{moving} / \Delta h_{stage}$
Blade speed	$u = \pi DN/60$
Axial thrust	$F_a = m\,(C_{f1} - C_{f2})$

3. Detailed Explanation

- **De Laval turbine:** single-stage impulse; very high speed (≈ 30,000 rpm), so speed must be reduced by gearing; for small outputs.
- **Velocity compounding (Curtis):** one set of nozzles followed by moving–fixed–moving blade rows; pressure drops once, velocity absorbed in stages.
- **Pressure compounding (Rateau):** several impulse stages in series, each with nozzles and blades; pressure drops in steps.
- **Axial thrust in reaction turbines:** the pressure difference across blades causes large axial thrust; balanced by a dummy piston and thrust bearing.

- **Turbine vs steam engine:** turbine — rotary, high speed, balanced, large capacity, little lubrication; steam engine — reciprocating, slow, unbalanced forces.

4. Comparison Table — Impulse vs Reaction Turbine

Basis	Impulse	Reaction
Pressure drop	Only in nozzles	In both fixed and moving blades
Pressure over moving blades	Constant	Decreases
Blade profile	Symmetrical	Aerofoil, asymmetric
Degree of reaction	0	0.5 (Parsons)
Axial thrust	Small	Large
Examples	De Laval, Curtis, Rateau	Parsons

5. Solved Numerical Problems

Example 1. In an impulse turbine the nozzle angle $\alpha = 20°$. Find the maximum blade efficiency and optimum speed ratio.

Solution:

1. $\eta_{max} = \cos^2 20° = 0.9397^2 =$ **88.3%**
2. $u/C_1 = \cos 20°/2 =$ **0.47**

Example 2. $C_1 = 600$ m/s, $\alpha = 20°$ — find the optimum blade speed of an impulse turbine.

Solution:

1. $u = C_1 \cos \alpha/2 = 600 \times 0.9397/2 =$ **281.9 m/s**

Example 3. Turbine wheel diameter 1 m and speed 3000 rpm. Find the blade speed.

Solution:

1. $u = \pi \times 1 \times 3000/60 =$ **157 m/s**

Example 4. In a stage $u = 150$ m/s, $C_{w1} + C_{w2} = 400$ m/s, flow 5 kg/s. Find the power.

Solution:

1. $W = 150 \times 400 = 60{,}000$ J/kg
2. $P = 5 \times 60{,}000 =$ **300 kW**

Example 5. In a Parsons turbine $\alpha = 20°$. Find the maximum blade efficiency.

Solution:

1. $\cos^2 20° = 0.883$
2. $\eta_{max} = 2 \times 0.883/(1 + 0.883) =$ **93.8%**

6. Frequently Asked Facts

- Impulse: pressure drop only in nozzles; reaction: in both blade rows.

- Impulse: $u/C_1 = \cos \alpha/2$, $\eta_{max} = \cos^2\alpha$.
- Parsons – 50% reaction; De Laval – simplest impulse.
- Curtis – velocity compounding; Rateau – pressure compounding.

7. Note (Confusing Facts)

- In impulse turbines only velocity (not pressure) falls over moving blades.
- The purpose of compounding is to reduce speed (rpm), not to raise efficiency.
- Parsons' optimum ratio is cos α, not cos α/2.

Remember: One line: R = 0 impulse, R = 0.5 Parsons. Assertion–Reason: (A) Gearing is used with De Laval turbines. (R) Their rotational speed is very high. — Both true; R explains A.

30 Most Important One-Liner Facts — Steam Turbines

1. Pressure drop only in nozzles — turbine?

— Impulse.

2. Pressure drop in both blades — turbine?

— Reaction.

3. Simplest impulse turbine?

— De Laval.

4. 50% reaction turbine?

— Parsons.

5. R for impulse?

— 0.

6. Purpose of compounding?

— Reduce speed.

7. Velocity compounding?

— Curtis.

8. Pressure compounding?

— Rateau.

9. Impulse optimum ratio?

— cos α/2.

10. Impulse max efficiency?

— $\cos^2\alpha$.

11. Parsons optimum ratio?

— cos α.

12. Parsons max efficiency?

— $2\cos^2\alpha/(1 + \cos^2\alpha)$.

13. Blade speed?

— $\pi DN/60$.

14. Work per kg?

— $u(C_{w1} \pm C_{w2})$.

15. Impulse blade profile?

— Symmetrical.

16. Reaction blade profile?

— Aerofoil.

17. More axial thrust?

— Reaction turbine.

18. Balancing axial thrust?

— Dummy piston.

19. Approx. speed of De Laval?

— 30,000 rpm.

20. Radial flow steam turbine?

— Ljungström.

21. Turbine supplying exhaust to process?

— Back-pressure.

22. Types of turbine governing?

— Throttle, nozzle control, bypass.

23. Degree of reaction?

— Δh moving ÷ Δh stage.

24. Moving blades act as nozzles — turbine?

— Reaction.

25. Impulse η_{max} at α = 20°?

— ≈ 88%.

26. Prime mover of thermal plant?

— Steam turbine.

27. Pressure drop in Curtis stage?

— Only once in nozzles.

28. Symbol of relative velocity?

— V.

29. Whirl velocity?

— Component along blade motion.

30. Advantage over steam engine?

— Balanced rotary motion.

Practice MCQs — Steam Turbines

Q1. In an impulse turbine the pressure drop occurs:

(a) Only in moving blades
(b) Only in nozzles
(c) In both
(d) Nowhere

✔ **Answer: (b) Only in nozzles**

Explanation: Impulse.

Q2. Parsons turbine is a:

(a) Impulse
(b) 50% reaction
(c) Velocity compounded
(d) Radial

✔ **Answer: (b) 50% reaction**

Explanation: R = 0.5.

Q3. Maximum blade efficiency of an impulse turbine:

(a) cos α
(b) $\cos^2\alpha$
(c) cos α/2
(d) $2\cos^2\alpha/(1 + \cos^2\alpha)$

✔ **Answer: (b) $\cos^2\alpha$**

Explanation: At $u/C_1 = \cos\alpha/2$.

Q4. Curtis turbine is:

(a) Pressure compounded
(b) Velocity compounded
(c) Reaction
(d) Radial

✔ **Answer: (b) Velocity compounded**

Explanation: Curtis.

Q5. Main purpose of compounding:

(a) Increase efficiency
(b) Reduce rotor speed
(c) Increase pressure
(d) Reduce weight

✔ **Answer: (b) Reduce rotor speed**

Explanation: Avoid excessive rpm.

Q6. Optimum u/C_1 for an impulse turbine:

(a) cos α
(b) cos α/2
(c) 2cos α
(d) sin α

✔ **Answer: (b) cos α/2**

Explanation: cos α/2.

Q7. Pressure across moving blades in a reaction turbine:

(a) Constant
(b) Decreases
(c) Increases
(d) Zero

✔ **Answer: (b) Decreases**

Explanation: Blades act as nozzles.

Q8. α = 20°, C_1 = 600 m/s — impulse optimum blade speed:

(a) 564 m/s
(b) 282 m/s
(c) 300 m/s
(d) 141 m/s

✔ **Answer: (b) 282 m/s**

Explanation: 600 × 0.94/2.

Q9. D = 1 m, N = 3000 rpm — blade speed:

(a) 50 m/s
(b) 157 m/s
(c) 314 m/s
(d) 3000 m/s

✔ **Answer: (b) 157 m/s**

Explanation: π × 1 × 50.

Q10. Blades of an impulse turbine are:

(a) Symmetrical
(b) Aerofoil
(c) Flat
(d) Curved asymmetric

✔ **Answer: (a) Symmetrical**

Explanation: Equal inlet and exit angles.

Q11. Largest axial thrust occurs in:

(a) Impulse
(b) Reaction
(c) De Laval
(d) Curtis

✔ **Answer: (b) Reaction**

Explanation: Pressure difference across blades.

Q12. Rateau turbine is:

(a) Velocity compounded
(b) Pressure compounded
(c) Reaction
(d) Single stage

✔ **Answer: (b) Pressure compounded**

Explanation: Rateau.

Q13. Degree of reaction of an impulse turbine:

(a) 0 (b) 0.5

(c) 1 (d) 0.25

✔ **Answer: (a) 0**

Explanation: No pressure drop in moving blades.

Q14. Speed of a De Laval turbine is about:

(a) 300 rpm (b) 3000 rpm

(c) 30,000 rpm (d) 300,000 rpm

✔ **Answer: (c) 30,000 rpm**

Explanation: Very high.

Q15. u = 150 m/s, ΣC_w = 400 m/s, 5 kg/s — power:

(a) 30 kW (b) 300 kW

(c) 3000 kW (d) 60 kW

✔ **Answer: (b) 300 kW**

Explanation: 5 × 150 × 400 W.

SSC JE / UPSSSC JE Previous Year Questions — Steam Turbines

Q1. De Laval turbine is a:

SSC JE (Previous Year)

(a) Simple impulse turbine (b) Reaction turbine

(c) Velocity compounded (d) Pressure compounded

✔ **Answer: (a) Simple impulse turbine**

Explanation: Single-stage impulse.

Q2. Example of a 50% reaction turbine:

SSC JE (Previous Year)

(a) Curtis (b) Parsons

(c) De Laval (d) Rateau

✔ **Answer: (b) Parsons**

Explanation: Parsons.

Q3. In an impulse turbine steam expands in:

SSC JE (Previous Year)

(a) Nozzles only (b) Moving blades only

(c) Both (d) Condenser

✔ **Answer: (a) Nozzles only**

Explanation: Nozzles.

Q4. Compounding in steam turbines reduces:

SSC JE (Previous Year)

(a) Efficiency
(b) Rotor speed
(c) Steam pressure
(d) Number of blades

✔ **Answer: (b) Rotor speed**

Explanation: Speed reduces.

Q5. Degree of reaction is defined as:

SSC JE (Previous Year)

(a) Enthalpy drop in moving blades ÷ total stage drop
(b) Fixed blade drop ÷ total
(c) Work ÷ heat
(d) Pressure ratio

✔ **Answer: (a) Enthalpy drop in moving blades ÷ total stage drop**

Explanation: Definition.

UPSSSC ALERT